

Penalty view

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$

Weighted costs

λ_1 λ_2 λ_K

$$\sum_k \lambda_k J_{c_k}(\pi)$$

Exact settings

$c = 0$
fixed β
fixed penalty
adaptive β
Lagrangian

Joint constraints

$$C_1 \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare \end{bmatrix} b_1$$

$$C_2 \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare \end{bmatrix} b_2$$

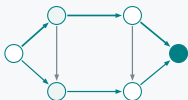
$$C_K \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare \end{bmatrix} b_K$$

Hard graph support



$$F_H = 0$$

Graph optimization



$$\min_{F \geq 0} -\langle A, F \rangle + \varepsilon KL(F \parallel \hat{F})$$

discounted flow balance

$$\langle C_k, F \rangle \leq b_k$$

Shared safety price

$$\lambda^* \rightarrow \text{Clip + filter}$$

$$\lambda_k^* \rightarrow \text{graph price} \quad \beta \rightarrow \text{cost feedback}$$

$$+$$

$$\lambda_k = \lambda_k^* + \beta_k$$

λ_1 λ_2 ... λ_K

One policy update

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO

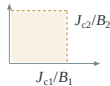
SafeOT-Dual

One shared price construction

Constraint view

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_c(\pi) \leq B$$

Joint budgets



Graph-price limits

under assumptions

$\varepsilon \rightarrow 0$:
LP-derived price
 $\varepsilon \rightarrow \infty$:
clipped active-set limit

(a) Rollout & graph

Policy rollout

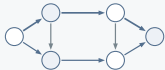
$s_0 \rightarrow s_1 \rightarrow \dots \rightarrow s_T$

rewards    

K costs    

↓ build graph

Empirical graph



$\hat{F}, A, C_k, \mu_0$

$\bar{J}_k$ (episode costs)

GAE advantages

(b) Unified price construction

Joint graph solve



$$b_k = B_k/L$$
$$F_H = 0$$

$$\min_{F \geq 0} -\langle A, F \rangle + \varepsilon KL(F \parallel \hat{F})$$

$$F1 - \gamma F^T 1 = (1-\gamma)\mu_0$$

$$\langle C_k, F \rangle \leq b_k \text{ (soft channels)}$$

Newton in u

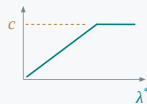
Bisection in λ

Episode-cost feedback

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+$$

Clip + filter

$$\lambda^- \leftarrow (1-\alpha)\lambda^-$$
$$+ \alpha \min(\lambda^*, c)$$



β

+

$$\lambda_k = \lambda_k^- + \beta_k$$

one price vector

(c) Policy update

Priced advantage

$$\tilde{A} =$$

$$\frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO
policy step

$$\pi_{t+1}$$

GAE: A_r, A_{c_k}

next rollout